Yuanye Lin

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Research interests: galaxy chemical evolution and the baryon cycle; galaxy environments and the cosmic web; spatially resolved galaxy evolution; Milky Way dust and interstellar gas.

Education

B.Sc. in Physics, Zhiyuan College, Shanghai Jiao Tong University 2017 – 2021

Ph.D. candidate in Astronomy, Shanghai Jiao Tong University 2021 – present

Expected Jun. 2027. Advisor: Prof. Ying Zu; Zhiyuan Honors Ph.D. Program.

Research Experience

Visiting Ph.D. Student, University of Utah, USA Sep. 2025 – Sep. 2026

Hosts: Profs. Zheng Zheng and Kyle Dawson, Department of Physics and Astronomy.

Grants and Fellowships

Principal Investigator, NSFC Young Student Basic Research Program May 2024 – Dec. 2026

Doctoral student program. Total funding: RMB 300,000.

"Reconstructing the Chemical Evolution of Galaxies across the Cosmic Web with DESI and JUST."

Project No. 123B240.

T. D. Lee Ph.D. Fellowship, Shanghai Jiao Tong University 2023 – 2025

Publications and Manuscripts

Refereed publications

1. **Lin, Y.**, & Zu, Y. 2023. "Constraints on galactic outflows from the metallicity–stellar mass–SFR relation of EAGLE simulation and SDSS galaxies." *MNRAS*, **521**, 411–432. doi:10.1093/mnras/stad502.

Submitted; public preprint available

2. **Lin, Y.**, Zu, Y., et al. 2026. "Satellite Metallicity Enhancement I: Suppressed Star Formation, Stellar Mass Loss, and Enriched Inflow of DESI and EAGLE Galaxies around Massive Clusters." Submitted to *ApJ*; response to referee comments in preparation. arXiv:2605.08342.

Unpublished manuscripts

3. **Lin, Y.**, et al. "Galactic Extinction and Reddening from DESI Luminous Red Galaxies." Under review by the DESI Collaboration.

4. **Lin, Y.**, Zu, Y., et al. "Satellite Metallicity Enhancement II: Stellar-Mass and Halo-Mass Dependence of Satellite Metallicity Enhancement in DESI DR1." Manuscript complete; DESI collaboration review in preparation.

Co-authored preprints

5. Shao, Z., Zu, Y., Salcedo, A., **Lin, Y.**, et al. 2026. "Extending the Stellar-to-Halo Mass Relation to Dwarf Galaxies with DESI DR1." arXiv:2609.04351.

Selected Research Contributions

Non-equilibrium chemical evolution model (NE-CEM)

- Formulate the model assumptions and derive the evolution equations, building on Ying Zu's model concept and EAGLE-to-SDSS strategy.
- Develop a power-law–exponential ("powexp") description of star formation histories and an empirical outflow mass-loading prescription using EAGLE, assembling a nine-parameter model that reconstructs its fundamental metallicity relation (FMR).
- Apply NE-CEM to reconstruct the SDSS FMR and use the inferred parameters to constrain the stellar mass–mass-loading relation, with comparisons to observational estimates.

Satellite metallicity enhancement (SME)

- Measure satellite metallicity enhancement with DESI and compare it with EAGLE. Construct three simulation reference samples to disentangle the contributing physical processes.
- Use NE-CEM to analyze how suppressed star formation, stellar mass loss, and enriched inflow contribute to satellite metallicity enhancement.
- Use three different metallicity calibrators in SME II to measure the dependence of the enhancement on halo mass, stellar mass, and star formation rate, and interpret the trends within the SME I framework.

Milky Way dust mapping with DESI luminous red galaxies

- Apply the "standard crayons" method of Peek & Graves (2010) to DESI LRGs, following Zheng Zheng's proposal. Construct the sample, validate the method, and develop the measurement pipeline.
- Apply three photometric corrections—airmass, Gaia zero-point, and filter-bandpass corrections—to place the LRG photometry on a common system and reduce systematic biases in the reddening measurements.
- Use D4000 and redshift to calibrate intrinsic LRG colors and construct a Milky Way reddening map across the DESI footprint. The map agrees with SFD and stellar-based reddening measurements on large angular scales; the method also enables measurements of R_V.
- Make initial foreground Na I D absorption-velocity measurements across the DESI footprint, using equivalent widths and velocities to investigate the column density and spatial distribution of gas associated with Milky Way dust.

Collaborations and Survey Projects

- Member, Dark Energy Spectroscopic Instrument (DESI) Collaboration.
- Lead, DESI Project #419: "A Panoramic View of the Chemical Evolution History of the DESI Y1 Galaxies at $z < 0.6$."
- Lead, DESI Project #759: "Measuring Galactic Reddening with DESI LRGs as Standard Crayons."
- Participant, Prime Focus Spectrograph (PFS) project.
- Contributor to the early survey design of the Jiao Tong University Spectroscopic Telescope (JUST).

Technical Skills

- **Software and computing:** Python, MATLAB, LaTeX; HPC/cluster computing.
- **Survey data and simulations:** DESI, SDSS; EAGLE, TNG.
- **Languages:** Chinese (native), English (fluent).